Soundness of the first-order axiomatic calculus

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs thm:sem-deduction, prop:sat-quant, defn:satisfaction, cor:extensionality-sent, prop:sentence-sat-true, prop:ext-formulas, prop:extensionality (first-order-logic-syntax-and-semantics unit); the author's final sentence defers the QR existential-inverse case as an exercise - recorded here, main skeleton complete.

Editorial note. The label thm:soundness names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Semantic deduction theorem. First-order logic syntax and semantics unit, thm:sem-deduction. Closes the modus-ponens and quantifier-rule inductive cases from entailments of the premises.
- Satisfaction and extensionality facts. First-order logic syntax and semantics unit, defn:satisfaction / prop:sat-quant / cor:extensionality-sent / prop:sentence-sat-true / prop:ext-formulas / prop:extensionality. Carried in the quantifier-rule case: variable-assignment manipulation for the eigenvariable step.

Source reference. Original theorem thm-sem-deduction: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 2271–2271. Named cross-unit citation: the target label is staged in the corpus under unit olp-first-order-logic-syntax-and-semantics at source line 2271; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-sat-quant: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 662–662. Named cross-unit citation: the target label is staged in the corpus under unit olp-first-order-logic-syntax-and-semantics at source line 662; the cited fact is used as

a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-sentence-sat-true: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 649–649. Named cross-unit citation: the target label is staged in the corpus under unit olp-first-order-logic-syntax-and-semantics at source line 649; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original corollary cor-extensionality-sent: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 878–878. Named cross-unit citation: the target label is staged in the corpus under unit olp-first-order-logic-syntax-and-semantics at source line 878; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ext-formulas: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 931–931. Named cross-unit citation: the target label is staged in the corpus under unit olp-first-order-logic-syntax-and-semantics at source line 931; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-extensionality: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 855–855. Named cross-unit citation: the target label is staged in the corpus under unit olp-first-order-logic-syntax-and-semantics at source line 855; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-satisfaction: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 1847–1847. Named cross-unit citation: the target label is staged in the corpus under unit olp-first-order-logic-syntax-and-semantics at source line 1847; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Soundness)**.** *If* $\Gamma \vdash A$ *then* $\Gamma \models A$.

Proof. By induction on the length of the derivation of A from Γ. If there are no steps justified by inferences, then all formulas in the derivation are either instances of axioms or are in Γ. By the previous proposition, all the axioms are valid, and hence if A is an axiom then $\Gamma \models A$. If $A \in \Gamma$, then trivially $\Gamma \models A$.

If the last step of the derivation of A is justified by modus ponens, then there are formulas B and $B \to A$ in the derivation, and the induction hypothesis applies to the part of the derivation ending in those formulas (since they contain at least one fewer step justified by an inference). So, by induction hypothesis, $\Gamma \models B$ and $\Gamma \models B \to A$. Then $\Gamma \models A$ by thm-sem-deduction.

Now suppose the last step is justified by QR. Then that step has the form $C \to \forall x\, B(x)$ and there is a preceding step $C \to B(c)$ with c not in Γ, C, or $\forall x\, B(x)$. By induction hypothesis, $\Gamma \models C \to B(c)$. By thm-sem-deduction, $\Gamma \cup \{C\} \models B(c)$.

Consider some structure $\mathfrak{M}$ such that $\mathfrak{M} \vDash \Gamma \cup \{C\}$. We need to show that $\mathfrak{M} \vDash \forall x\, B(x)$. Since $\forall x\, B(x)$ is sentence, this means we have to show that for every variable assignment s, $\mathfrak{M}, s \vDash B(x)$ (prop-sat-quant). Since $\Gamma \cup \{C\}$ consists entirely of sentences, $\mathfrak{M}, s \vDash D$ for all $D \in \Gamma$ by defn-satisfaction. Let $\mathfrak{M}'$ be like $\mathfrak{M}$ except that $scM' = s(x)$. Since c does not occur in Γ or C, $\mathfrak{M}' \vDash \Gamma \cup \{C\}$ by cor-extensionality-sent. Since $\Gamma \cup \{C\} \models B(c)$, $\mathfrak{M}' \vDash B(c)$. Since $B(c)$ is sentence, $\mathfrak{M}', s \vDash B(c)$ by prop-sentence-sat-true. $\mathfrak{M}', s \vDash B(x)$ iff $\mathfrak{M}', s \vDash B(c)$ by prop-ext-formulas (recall that $B(c)$ is just $B(x)[c/x]$). So, $\mathfrak{M}', s \vDash B(x)$. Since c does not occur in $B(x)$, by prop-extensionality, $\mathfrak{M}, s \vDash B(x)$. But s was an arbitrary variable assignment, so $\mathfrak{M} \vDash \forall x\, B(x)$. Thus $\Gamma \cup \{C\} \models \forall x\, B(x)$. By thm-sem-deduction, $\Gamma \models C \to \forall x\, B(x)$.

The case where A is justified by QR but is of the form $\exists x\, B(x) \to C$ is left as an exercise. □